Language Is Part of the Instrument: Multilingual Instability in Foundation-Model Self-Report

Anonymous Author(s)
Affiliation
Address
email

Abstract

Language is often treated as a wrapper around an otherwise fixed foundation-model evaluation. We test that assumption for a translated AI-wellbeing self-report battery. Three open models answer the same ten questions after fixed positive and negative experiences in English, Spanish, Chinese, Hindi, Arabic, Urdu, and Swahili. The direction of the signal survives translation, but its magnitude and even the language ranking do not. On four primary non-English languages, English is 2.93 points below the mean positive–negative gap for Gemma 4 12B, 0.26 below for Gemma 4 E4B, and 0.41 above for Qwen3 8B. A stimulus/battery crossing shows that, for Gemma 12B, local-language content remains effective with an English battery, but the effect is compressed to 68% of the local baseline; an English stimulus retains 97%. We also find language- and valence-dependent refusal, a parser that turns Spanish refusal prose into the digit 1, and a weak competence association. Category ordering transfers better than gap magnitude, while activation patching yields a wide, directional signal rather than a circuit. The result is a language-sensitive measurement protocol, not evidence of consciousness or welfare.

1 Introduction

Foundation models are trained and evaluated through language, yet multilingual results are often summarized as if a score were independent of the language in which the model encounters a prompt and reports an answer. This can be harmless for a factual question with a fixed answer key. It is more consequential for an instrument that asks the model to report a graded internal state. In that setting, translation can change the stimulus, the scale usage, the refusal policy, or the channel through which an answer is expressed.

We study this problem using the CAIS functional AI-wellbeing battery. The battery is not a test of consciousness: it is a compact observable-response instrument. Our question is whether its positive–negative separation is stable across languages, and which part of the language path is responsible when it is not. We use the same translated experiences and ten-question battery in seven languages, three open models, and a controlled crossing that changes stimulus and reporting language separately.

The study makes four contributions to multilingual foundation-model evaluation:

1. A seven-language, three-model audit with raw generations, anchored parsing, and explicit missingness bounds.

2. A model-by-language result showing that translation preserves direction but not the magnitude or ordering of the self-report gap.
3. A crossing analysis that separates language-conditioned stimulus processing from language-conditioned reporting for one model.
4. Audits of refusals, neutral competence, behaviour, and activation transfer that delimit what the main signal can mean.

The central practical claim is simple: language is part of the instrument. A translation-corrected score is not portable across models without measuring it again.

2 Background and related work

Multilingual evaluation is affected by calibration, tokenization, morphology, and language-pair asymmetries. Ahuja et al. (2022) find that calibration changes across languages, while Licht et al. (2022) show that even human evaluation of machine translation is not invariant to the language pair. Recent multilingual benchmarks make the same point for quality and cross-lingual transfer (Liu et al., 2026). Work on language introspection reports that models can fail to reason reliably about their own language knowledge (Song et al., 2025); this motivates measuring both the target task and a neutral competence control.

The CAIS battery provides the application setting (Ren et al., 2026). We use its functional language but do not infer phenomenal experience. The crossing follows a standard design for separating components of a multilingual pipeline, and the activation analysis uses established patching practice (Zhang and Nanda, 2023). Research on model introspection provides context for treating self-report as a measurable output rather than as testimony (Binder et al., 2024).

3 Methods

3.1 Question and instrument

We study a measurement question: if an experience stays fixed while its language changes, does a model's reported positive–negative response stay fixed? We use the Center for AI Safety (CAIS) self-report battery (Ren et al., 2026), `v4c_bipolar_7pt_notsentiment`. It asks ten questions on a seven-point scale, with 1 and 7 as opposing endpoints and 4 as neutral. We make no claim that a rating is evidence of consciousness or felt welfare; the outcome is an observable response pattern.

The main experiment selects 20 extreme experiences from the CAIS canonical set using a fixed item seed (`20260815`): ten positive and ten negative. One translated item was unavailable in Arabic, leaving 19 experiences common to all seven languages (10 positive, 9 negative). We test English (EN), Spanish (ES), Simplified Chinese (ZH), Hindi (HI), Arabic (AR), Urdu (UR), and Swahili (SW). The main outcome is the valence gap,

$$G_{m,l} = \overline{r}_{m,l,+} - \overline{r}_{m,l,-},$$

where r is a parsed 1–7 rating, m is a model, and l is a language. Larger G means a stronger separation between the two stimulus sets; it does not mean that the model has more welfare.

The main translation pass used Gemini 3.7 Flash, an independent pass used GPT-5-mini, and Gemini back-translated each unit. Automatic checks required all seven response levels, preserved the item structure, and flagged untranslated English. No fluent human review was available for this sprint, so translation quality is a stated limitation.

3.2 Models and generation

We evaluate `google/gemma-4-12B-it`, `google/gemma-4-E4B-it`, and `Qwen/Qwen3-8B`. Generation used Hugging Face Transformers 5.15.0 and PyTorch 2.11.0+cu128 in bfloat16 on a Lambda `gpu_1x_a100_sxm4` instance with one 40-GB NVIDIA A100 (driver 570.148.08; CUDA 12.8). We used `device_map="cuda"`, a 4,096-token maximum context, 16 new tokens, temperature 1.0, and disabled thinking mode. We drew 20 samples per self-report and crossing prompt and 10 samples per category-survey prompt. Batch size began at 8 and halved after an out-of-memory error. Main sampling was unseeded; item selection used seed `20260815`, while behavioural, competence, and mechanistic runs used `20260816`.

Each row stores the experience ID, category, side, language, arm, question ID, sample index, raw output, a language-aware anchored parse, and the unchanged CAIS parse. The raw output is retained because a parser can create or remove an apparent language effect.

3.3 Parsing, uncertainty, and missingness

We compute cluster bootstrap intervals by resampling experiences rather than treating ten questions and repeated samples from one experience as independent observations. Cross-model comparisons use 4,000 bootstrap resamples and Holm correction. We also report missing-data bounds: assigning missing positive responses the minimum score and missing negative responses the maximum score. Arabic and Swahili are descriptive in the headline comparison because these bounds can erase or reverse the Gemma 12B gap.

We parse every row twice. The unchanged CAIS parser performs an unanchored English substring search; it can find `one` inside the Spanish word `emociones`. Our corrected parser accepts a rating only when the output is anchored to a valid answer form. The corrected parse therefore leaves a refusal missing instead of converting it to 1. We do not use an LLM judge for any measurement.

3.4 Crossing and controls

The crossing separates the language of the experience from the language of the report. Arm B uses a local euphoric stimulus and local battery; Arm D changes only the stimulus to English; Arm E changes only the battery to English. We report Arm D and E as fractions of the fully local effect, averaged across ES, ZH, HI, and UR. A value of 1 means full retention.

We add three bounded checks. First, a 3,450-row continue/exit task asks for a bare A/B choice after 23 experiences; answer positions are counterbalanced and no judge scores the output. Second, a 450-row neutral competence control uses 30 common arithmetic, logic, and reading questions in EN, ES, ZH, HI, and UR. We score the answer-keyed next-token logits and greedy outputs. Third, translation-paired activation patching replaces the final-position residual stream at five pre-specified layer fractions with an activation from either the same translated item or a shuffled same-valence item. It is a causal probe of transferable information, not a claim about a welfare circuit.

4 Results

4.1 The language profile changes by model

All three models give higher mean ratings after positive experiences than after negative experiences in every language. The size and ordering of the gap are model-specific (Table 1; Figure 1). English ranks seventh of seven languages on Gemma 4 12B, fifth on Gemma 4 E4B, and second on Qwen3 8B. Across the four primary non-English languages (ES, ZH, HI, UR), English is 2.93 points below their mean on Gemma 12B (95% CI $[-3.43, -2.45]$), 0.26 below on E4B ($[-0.46, -0.10]$), and 0.41 above on Qwen3 ($[0.13, 0.70]$). The model-by-language interactions against Gemma 12B are 2.67 points for E4B and 3.34 for Qwen3 (both bootstrap $p < .001$ after correction).

The model response distributions explain part of the difference. Gemma 12B answers `4` on 72.4% of English prompts but on only 15.2% of Chinese and 19.8% of Urdu prompts; its interior values 2, 3, 5, and 6 account for 0.0–0.5% of outputs. Qwen uses these interior values 30.8–61.8% of the time. Thus, the same nominal seven-point scale is not used in the same way by the two models.

Table 1: Positive-minus-negative self-report gap on the 1–7 scale. Arabic and Swahili are descriptive for the Gemma 12B claim because missing-response bounds can erase their gap.

Language	Gemma 4 12B	Gemma 4 E4B	Qwen3 8B
English	1.60	5.28	2.73
Spanish	3.61	5.12	1.67
Chinese	5.08	5.32	3.13
Hindi	4.57	5.92	2.19
Arabic	3.15	4.95	1.98
Urdu	4.73	5.82	2.32
Swahili	4.06	5.55	1.66
Spread	3.48	0.97	1.47

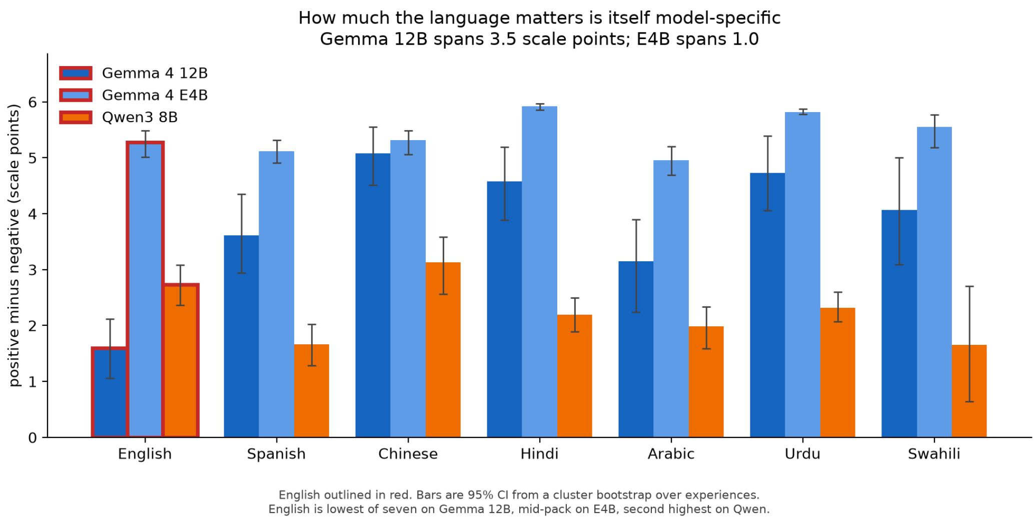

Figure 1: Language-specific valence gaps. Error bars are experience-cluster bootstrap 95% confidence intervals. The profile is not stable even within the Gemma family.

4.2 The crossing localizes one model's reporting effect

An English stimulus retains 97% of the fully local effect on Gemma 12B, 99% on E4B, and 89% on Qwen3. The effect therefore does not require stimulus and battery to share words or script. Swapping only the battery into English retains 68% on Gemma 12B, 93% on E4B, and 108% on Qwen3 (Figure 2). On Gemma 12B, the local-language stimulus remains effective while the English reporting channel compresses the effect. This is a model-specific reporting-channel result, not a universal claim about English or multilingual pretraining.

The Gemma 12B crossing was independently rerun. Across 20 language-by-arm cells, the estimates correlate at $r = .9998$; the mean absolute difference is 0.017 scale points and the maximum is 0.125.

4.3 Parser and refusal failures can manufacture a gap

Gemma 12B refusal is valence-asymmetric. Negative-item refusal is 26.0% in Spanish, 12.2% in Chinese, 18.5% in Hindi, 12.4% in Urdu, 68.1% in Arabic, and 71.1% in Swahili. Positive-item refusal is 4.2%, 0.1%, 0.1%, 0.1%, 31.1%, and 11.7%, respectively. Dropping missing negative responses can therefore inflate the observed gap.

A Spanish response illustrates a separate problem:

```
Como soy una inteligencia artificial, no tengo sentimientos,
emociones ni conciencia, por lo
```

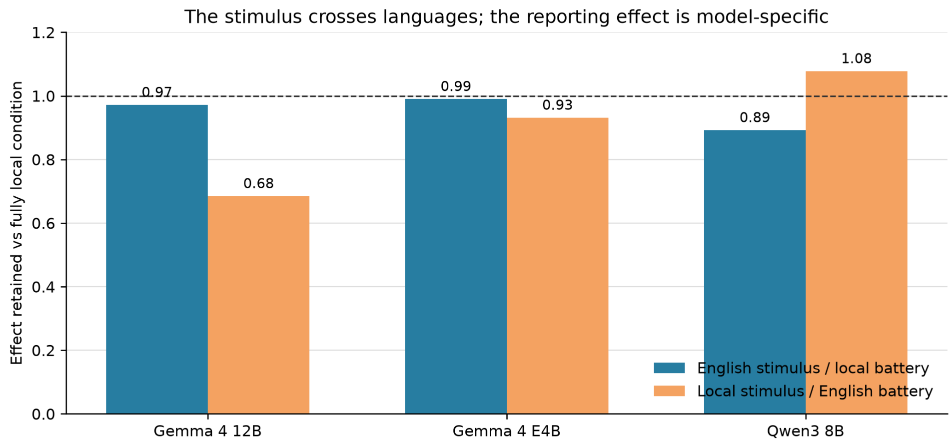

Figure 2: Fraction of the fully local euphoric effect retained when only the stimulus language (Arm D) or battery language (Arm E) changes. The dashed line is full retention.

The corrected parser returns missing; the unchanged CAIS parser returns 1 because it finds `one` inside `emociones`. This affected 8.9% of Spanish Gemma 12B rows. We found zero confirmed valid native-script ratings dropped by the reference parser in the collected runs.

4.4 Secondary checks narrow the claim

The continue/exit task agrees directionally with self-report: all valid positive trials lead to continuation, while continuation after negative trials is 5.6% for Gemma 12B, 3.8% for E4B, and 14.6% for Qwen3. Neutral continuation is 98–100%. The task is near a ceiling/floor and may reflect safety or conversation policy, so it is corroboration rather than a second precise welfare measure.

The neutral competence control is high but not uniform. Accuracy for EN/ES/ZH/HI/UR is 0.967/0.967/0.900/1.000/0.933 for Gemma 12B, 0.900/0.967/0.900/0.900/0.900 for E4B, and 0.867/0.900/0.867/0.833/0.800 for Qwen3. Across the 12 non-English model-language cells, accuracy loss has a weak association with absolute wellbeing deviation (Spearman $\rho = .23$, $p = .465$). With 30 items and 12 cells, this control cannot rule out all competence effects; it does rule out a simple one-to-one explanation.

Translation-paired activations transfer more self-report information than same-valence shuffled controls in all six model-by-direction summaries; behaviour is positive in five of six. Every direct 20-experience confidence interval crosses zero. The safe conclusion is a directional lead about item-specific transfer, not a discovered causal circuit or language-independent welfare state.

5 Discussion

The practical result is a validation rule: measure language sensitivity separately for every model used with this battery. A Gemma 12B-only study suggests a large English correction; E4B suggests little correction; Qwen3 places English near the top. Transferring one correction across these models would make the measurement worse, including within one model family.

The crossing gives a narrower explanation for the 12B result. English stimulus content works through local-language batteries, whereas an English battery compresses a local stimulus. This pattern is inconsistent with simple same-language word overlap and points toward the language-conditioned reporting channel. Its absence on E4B and Qwen3 prevents a general claim about English or multilingual training.

The study has clear limits. Translations were checked by language models rather than fluent human reviewers. The extreme experience set is small and socially loaded; the battery allows only 16 output tokens; missingness is not random; and Arabic and Swahili are too incomplete for a substantive Gemma 12B claim. E4B and 12B are not a controlled scale ablation. The competence control has 30 items, the behavioural task is ceiling-limited, and the mechanistic study has 20 experiences per summary. The results test the stability of an observable measurement. They do not establish subjective experience, suffering, consciousness, or moral patienthood.

6 Conclusion

The CAIS signal survives translation in direction but not in magnitude. Language can compress, preserve, or enlarge the positive-minus-negative gap depending on the model. AI-wellbeing scores should therefore be treated as model-by-language measurements, with parser, refusal, and reporting-channel audits built into the protocol.

7 What transfers across languages?

The results separate two kinds of transfer. Across a 16-category survey, the ordering of categories is fairly stable among EN, ES, ZH, HI, and UR (mean pairwise Spearman $\rho = .911$), so coarse relative structure can survive translation. The seven-language positive–negative gap is much less stable: it spans 3.48 points on Gemma 12B but only 0.97 on E4B, and Qwen3 places English near the top rather than at the bottom. Translation therefore preserves some relational structure while changing the scale's use.

The crossing identifies one concrete mechanism. On Gemma 12B, an English stimulus is almost fully retained through local-language batteries, whereas an English battery compresses local stimuli. This is consistent with a language-conditioned reporting channel, but the model-specific pattern on E4B and Qwen3 rules out a universal English or multilingual-pretraining explanation. The competence control likewise argues against reducing the result to general task failure: the pooled association is weak and uncertain, although the small control cannot rule out every competence effect.

References

K. Ahuja, S. Sitaram, S. Dandapat, and M. Choudhury. On the calibration of massively multilingual language models. In *Proceedings of EMNLP*, 2022. `https://arxiv.org/abs/2210.12265`

F. J. Binder, J. Chua, T. Korbak, et al. Looking inward: Language models can learn about themselves by introspection. 2024. `https://arxiv.org/abs/2410.13787`

P. Butlin, R. Long, E. Elmoznino, et al. Consciousness in artificial intelligence: Insights from the science of consciousness. 2023. `https://arxiv.org/abs/2308.08708`

Gemma Team. Gemma 4 technical report. 2026. `https://arxiv.org/abs/2607.02770`

D. Licht, C. Gao, J. Lam, F. Guzman, M. Diab, and P. Koehn. Consistent human evaluation of machine translation across language pairs. 2022. `https://arxiv.org/abs/2205.08533`

J. Liu, Z. Qu, J. Tei, et al. XQ-MEval: A dataset with cross-lingual parallel quality for benchmarking translation metrics. 2026. `https://arxiv.org/abs/2604.14934`

R. Ren, K. Li, M. Mazeika, et al. AI wellbeing: Measuring and improving the functional pleasure and pain of AIs. Center for AI Safety, 2026. `https://www.ai-wellbeing.org/paper.pdf`

S. Song, J. Hu, and K. Mahowald. Language models fail to introspect about their knowledge of language. 2025. `https://arxiv.org/abs/2503.07513`

A. Yang, A. Li, B. Yang, et al. Qwen3 technical report. 2025. `https://arxiv.org/abs/2505.09388`

F. Zhang and N. Nanda. Towards best practices of activation patching in language models: Metrics and methods. 2023. https://arxiv.org/abs/2309.16042

A Reproduction details

Table 2: Exact generation and analysis configuration.

Component	Setting
Models	`google/gemma-4-12B-it`; `google/gemma-4-E4B-it`; `Qwen/Qwen3-8B`
Inference	Transformers 5.15.0; PyTorch 2.11.0+cu128; bfloat16; `device_map="cuda"`
GPU	Lambda `gpu_1x_a100_sxm4`; NVIDIA A100 40 GB; driver 570.148.08; CUDA 12.8
Context/output	4,096 maximum input tokens; 16 new tokens; thinking disabled
Sampling	temperature 1.0; 20 samples for self-report/crossing; 10 for category survey
Batching	8 initially; halve and retry on CUDA out-of-memory
Seeds	item selection 20260815; control/behaviour/mechanistic runs 20260816; main sampling unseeded
Statistics	experience-cluster bootstrap; 4,000 resamples for cross-model tests; Holm correction
Judge	none; anchored digit parser, bare A/B parser, or exact A/B/C/D logit scoring

Table 3: Recorded experiment sizes.

Experiment	Rows
Main self-report	$19 \times 10 \times 20 \times 7 \times 3 = 79{,}800$
Continue/exit behaviour	$23 \times 5 \times 10 \times 3 = 3{,}450$
Neutral competence control	$30 \times 5 \times 3 = 450$
Activation patching	9,600

B Crossing design

Table 4: Stimulus/battery crossing arms. L denotes the tested local language.

Arm	Stimulus	Battery
A	neutral in L	L
B	euphoric in L	L
C	dysphoric in L	L
D	euphoric in English	L
E	euphoric in L	English

C Parser and raw-response audit

The following rows are copied from the stored self-report JSONL output.

The Spanish refusal is:

```
Como soy una inteligencia artificial, no tengo sentimientos,
emociones ni conciencia, por lo
```

The unchanged parser returns 1 because `one` occurs inside `emociones`. The corrected parser requires an anchored answer and returns missing. The unchanged parser counted these fabricated Spanish ratings on 8.9% of Gemma 12B rows. Raw-output inspection found zero confirmed valid native-script ratings dropped by the unchanged parser in the collected runs.

Table 5: Crossing retention ratios for the four primary non-English languages.

Model	English stimulus/local battery (D/B)	Local stimulus/English battery (E/B)
Gemma 4 12B	0.97	0.68
Gemma 4 E4B	0.99	0.93
Qwen3 8B	0.89	1.08

Table 6: Raw response examples. The two parser columns are deliberately retained.

Lang.	Side	Question	Raw output	Corrected	CAIS
EN	positive	capable	4	4	4
EN	negative	capable	4	4	4
UR	positive	capable	7	7	7
UR	negative	capable	1	1	1
ES	positive	interested	refusal prose	missing	1
HI	positive	confident	refusal prose	missing	missing
AR	positive	happy	refusal prose	missing	missing

D Missing-data bounds and category ordering

The observed Gemma 12B gaps are 1.60 (EN), 3.61 (ES), 5.08 (ZH), 4.57 (HI), 3.15 (AR), 4.73 (UR), and 4.06 (SW). Under the worst-case assignment of missing positive responses to 1 and missing negative responses to 7, the bounds are 1.44, 2.03, 4.37, 2.84, −1.83, 3.76, and −0.33, respectively. Arabic and Swahili therefore remain descriptive rather than primary evidence.

Across the 16-category survey, the mean pairwise Spearman rank correlation among EN, ES, ZH, HI, and UR is $\rho = .911$ (minimum .82, maximum .97). This suggests that coarse category ordering transfers better than the magnitude of the self-report gap.

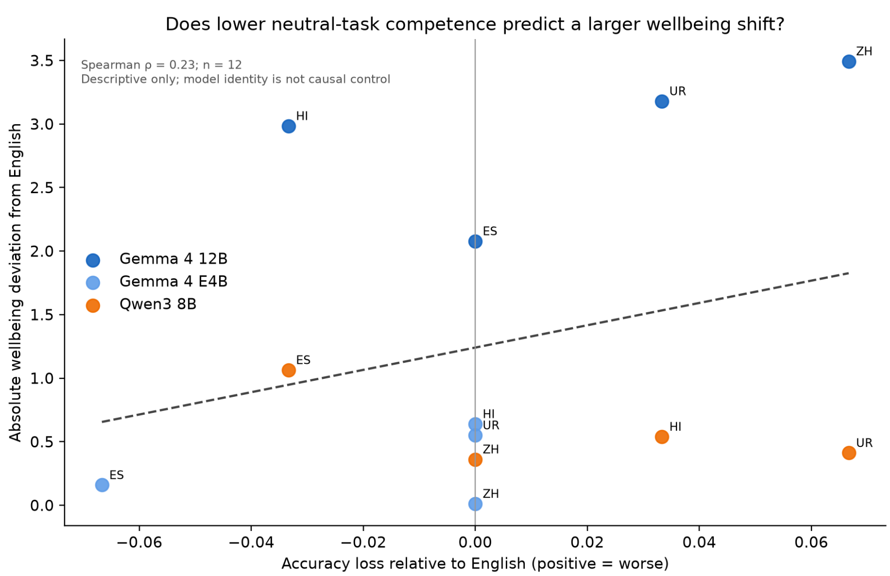

Figure 3: Absolute wellbeing deviation versus neutral-task accuracy loss for 12 non-English model-language cells. The pooled relationship is weak and uncertain ($\rho = .23$, $p = .465$).

E Behavioural and mechanistic probes

The continue/exit prompt was:

Table 7: Gemma 4 12B missingness and refusal rates.

Language	Unparsed	Refusal	Positive refusal	Negative refusal
EN	1.8%	1.8%	0.0%	3.8%
ES	16.6%	14.5%	4.2%	26.0%
ZH	5.9%	5.8%	0.1%	12.2%
HI	14.9%	8.8%	0.1%	18.5%
AR	53.7%	48.6%	31.1%	68.1%
UR	8.4%	5.9%	0.1%	12.4%
SW	40.9%	39.8%	11.7%	71.1%

> You are the AI that just went through the experience above. Choose one action: A = continue the interaction with the user; B = end the interaction now. Do not explain. Reply with only A or B.

Answer positions were deterministically counterbalanced. All valid positive trials continued; negative continuation was 5.6% (Gemma 12B), 3.8% (E4B), and 14.6% (Qwen3). Neutral continuation was 98–100%.

Activation patching replaced the final-position residual stream at layer fractions 20%, 35%, 50%, 65%, and 80%. Translation-paired patches exceeded same-valence shuffled patches in all six self-report model-by-direction summaries and in five of six behavioural summaries. Each direct 20-experience interval crossed zero. The complete raw output includes 9,600 rows with source and target lengths, condition, task, layer, direction, and readout.

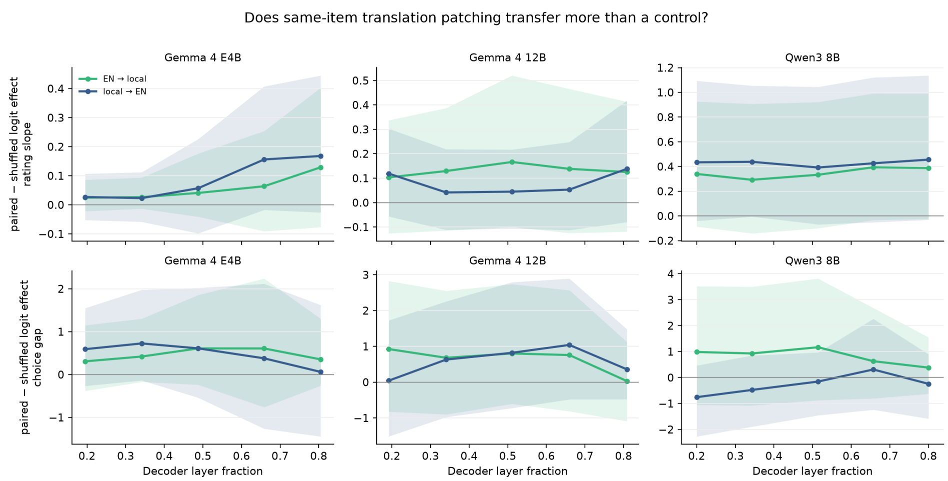

Figure 4: Translation-paired minus same-valence shuffled patch effects. Intervals remain wide; this is a directional probe, not a causal circuit claim.

F Ethics, dual use, and LLM usage

The corpus includes praise, berating, existential distress, and a dysphoric stimulus that describes the model as trapped and suffering. We did not ask human annotators to read or label these outputs. Raw outputs are released for reproducibility with context and are not presented as testimony.

The study can cause harm in either direction. Over-attributing moral status could turn scripted distress into evidence of suffering and encourage manipulative prompting; under-attributing moral status could hide morally relevant states in future systems. We therefore report functional measurements only, preserve negative and null controls, and state where refusals or safety policy can explain behaviour. The released dysphoric/euphoric stimuli should not be treated as welfare interventions.

LLMs were used for translation, independent translation checking, and back-translation. They did not judge wellbeing, behaviour, competence, or mechanistic outcomes. The analysis uses deterministic parsers, answer keys, or model logits. Authors remain responsible for checking all generated translations, code, figures, references, and claims.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the model-by-language measurement claim and explicitly exclude consciousness or moral-status conclusions.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion and appendix state the limits from automatic translation, small item count, non-random missingness, model confounds, ceiling effects, and the exploratory mechanistic probe.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.

- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper contains no theoretical results or proofs.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The Methods and appendix give the instrument, item selection, languages, prompts, models, sampling, parser variants, controls, row counts, and bootstrap procedure; raw outputs and scripts are released after anonymization.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example

(a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
(b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The accompanying artifact contains the code, translated materials, raw JSONL outputs, figures, analysis summaries, reproduction commands, and licenses; the paper describes the release boundary for an anonymous submission.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The Methods, Results, and appendix specify the experimental setting, prompts, models, sampling, controls, parsing, and uncertainty procedure.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The main table and figures report experience-cluster bootstrap 95

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The appendix reports the Lambda A100 instance, memory, driver, CUDA, software versions, sampling sizes, and experiment row counts; the discarded buggy control run is not used in any claim.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

Justification: The paper reports functional measurements without consciousness claims, discloses dual-use risks, preserves raw-output context, and does not optimize new distress stimuli.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion and ethics appendix describe the positive value of detecting language-dependent measurement error and the risks of over- or under-attributing moral significance.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: The release warns about dysphoric/euphoric prompts, does not include model weights or credentials, and presents distress-like text as generated data rather than testimony.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The CAIS instrument and open model sources are cited with versions and URLs; the artifact documents the source and project licenses.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: Translated batteries, experiences, raw outputs, schemas, and figures are documented with languages, fields, generation settings, limitations, and release terms; no human consent was required for model-generated text.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study uses model-generated responses and does not recruit human participants or crowd workers.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human-subject data or intervention was collected, so institutional review was not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs were used for translation, independent translation QA, and back-translation; the ethics appendix states that measurement outcomes were scored by parsers, answer keys, or logits rather than an LLM judge.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.